

Yanxiao Liu

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Background

Imperial College London, 帝国理工学院 2025.10 –

Postdoctoral Research Associate, Project: [Machine Learning and Information Theory]

Advisor: Prof. Deniz Gündüz (IEEE Fellow)

The Chinese University of Hong Kong, 香港中文大学 2021.8 – 2025.7

PhD, Department of Information Engineering

Advisors: Prof. Cheuk Ting Li and Prof. Raymond W. Yeung (IEEE Life Fellow)

GPA: 3.794/4.000

Stanford University 2024.1 – 2024.6

Visiting Researcher, Department of Electrical Engineering

Advisor: Prof. Ayfer Özgür

The Chinese University of Hong Kong, Shenzhen, 香港中文大学（深圳） 2017 – 2021

B.Eng. in Electronic Information Engineering

Advisor: Prof. Shenghao Yang

Research

I am an information theorist with broad interests in computer science and engineering, including information processing, inference, privacy, security, and distributed machine learning.

- **Information Theory and Communications:** I provide nonasymptotic coding schemes, low-complexity algorithms and theoretic analyses for fundamental problems in information theory.
- **Compression and Privacy for Distributed Learning:** I employ information-theoretic techniques (specifically, *remote source generations*) to compress privacy mechanisms, and solve various canonical tasks in distributed analytics and federated learning.
- **Security and Watermarking:** I design novel watermarking schemes that are distortion-free to original texts and distributionally-robust against attacks, applicable to effective protection and detection of content from generative AI models.

Selected Awards

- Chinese Institute of Electronics Information Theory Annual Conference Paper Award, 2025
 - 中国电子学会信息论年会（2025）最佳论文奖
- CUHK PhD International Mobility for Partnerships and Collaborations Award 2024 ($\approx 1.5\%$).
- NeurIPS Scholar Award (2024).
- ISIT Travel Award (2024).

Publications

Journal Articles

- [1] **Yanxiao Liu**, Sepehr Heidari Advary, and Cheuk Ting Li. “Nonasymptotic Oblivious Relaying and Variable-Length Noisy Lossy Source Coding”. In: *IEEE Transactions on Information Theory*. 72.6 (2026).
- [2] **Yanxiao Liu** and Cheuk Ting Li. “One-Shot Coding over General Noisy Networks”. In: *IEEE Transactions on Information Theory*. 71.11 (2025), pp. 8346–8357.
- [3] Chih Wei Ling*, **Yanxiao Liu*(co-first)**, and Cheuk Ting Li. “Weighted Parity-Check Codes for Channels with State and Asymmetric Channels”. In: *IEEE Transactions on Information Theory*. 70.8 (2024), pp. 5573–5588.
- [4] Shenghao Yang, Jun Ma, and **Yanxiao Liu**. “Wireless Network Scheduling with Discrete Propagation Delays: Theorems and Algorithms”. In: *IEEE Transactions on Information Theory*. 70.3 (2024), pp. 1852–1875.

Conference Proceedings

- [5] **Yanxiao Liu**, Wei-Ning Chen, Ayfer Özgür, and Cheuk Ting Li. “Universal Exact Compression of Differentially Private Mechanisms”. In: *The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2024.
- [6] **Yanxiao Liu**, Shenghao Yang, and Cheuk Ting Li. “Joint Scheduling and Multiflow Maximization in Wireless Networks”. In: *2026 International Zurich Seminar on Information and Communication (IZS)*. 2026.
- [7] **Yanxiao Liu**, Chun Hei Michael Shiu, Lele Wang, and Deniz Gündüz. “On the Generalization Error of Differentially Private Algorithms Via Typicality”. In: *2026 IEEE International Symposium on Information Theory (ISIT)*. 2026.
- [8] **Yanxiao Liu**, Sepehr Heidari Advary, and Cheuk Ting Li. “Nonasymptotic Oblivious Relaying and Variable-Length Noisy Lossy Source Coding”. In: *2025 IEEE International Symposium on Information Theory (ISIT)*. 2025.
- [9] **Yanxiao Liu** and Cheuk Ting Li. “One-Shot Information Hiding”. In: *2024 IEEE Information Theory Workshop (ITW)*. 2024.
- [10] Yijun Fan, **Yanxiao Liu**, Yi Chen, Shenghao Yang, and Raymond W. Yeung. “Reliable Throughput of Generalized Collision Channel Without Synchronization”. In: *2023 IEEE International Symposium on Information Theory (ISIT)*. 2023.
- [11] Yijun Fan, **Yanxiao Liu**, and Shenghao Yang. “Continuity of Link Scheduling Rate Region for Wireless Networks with Propagation Delays”. In: *2022 IEEE International Symposium on Information Theory (ISIT)*. 2022.
- [12] Chih Wei Ling, **Yanxiao Liu**, and Cheuk Ting Li. “Weighted Parity-Check Codes for Channels with State and Asymmetric Channels”. In: *2022 IEEE International Symposium on Information Theory (ISIT)*. 2022.
- [13] Jun Ma, **Yanxiao Liu**, and Shenghao Yang. “Rate Region of Scheduling a Wireless Network with Discrete Propagation Delays”. In: *IEEE INFOCOM 2021-IEEE Conference on Computer Communications (INFOCOM)*. 2021.

Professional Services

Award

- Top Reviewer for NeurIPS 2025 ($\approx 8\%$).

TPC member

- IEEE GLOBECOM 2026

Session Chair

- Information Theory Workshop 2024.

Reviewer

- Annual Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Machine Learning (ICML)
- IEEE Transactions on Information Theory (TIT)
- IEEE Transactions on Automatic Control (TAC)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Journal on Selected Areas in Communications (JSAC)
- IEEE International Symposium on Information Theory (ISIT)
- IEEE International Conference on Communications (ICC)
- Data Compression Conference (DCC)

Industrial Experience

Shenzhen Research Institute of Big Data 2021.5–2021.8
Research Intern

Pingan Technology 2020.8–2020.10
Software Engineer Intern

References

Cheuk Ting Li PhD Advisor
Associate Professor The Chinese University of Hong Kong

Raymond W. Yeung PhD Advisor
Choh-Ming Li Professor The Chinese University of Hong Kong

IEEE Life Fellow

Deniz Gündüz Postdoc Advisor
Professor Imperial College London

IEEE Fellow

Ayfer Özgür Visiting Host
Professor Stanford University

Vincent Y. F. Tan Thesis External Examiner
Professor National University of Singapore

Chandra Nair Thesis Committee Chair
Professor The Chinese University of Hong Kong

IEEE Fellow